

O & Soergel, lec 3: Overview III

- 1) Toward Koszul duality
- 2) Geometric interpretation

1) Toward Koszul duality

In Sec 1.2 of Lec 2 we produced an equivalence

$\mathcal{O}^{\text{pr-proj}} \xrightarrow{\sim} S\text{Mod}_{\text{agr}}(W)$. This yields an algebra isom'm

$$(1) \quad A := \text{End}\left(\bigoplus_{w \in W} P_0(w \cdot 0)\right) \xrightarrow{\sim} \text{End}_{K[y^*]}\left(\bigoplus_{w \in W} B_w \otimes_{K[y^*]} K_0\right)$$

$M := \bigoplus_{w \in W} B_w \otimes_{K[y^*]} K_0$ is a fin. dim. graded $K[y^*]$ -module

So the r.h.s. of (1) is canonically graded: if $M = \bigoplus_{j \in \mathbb{Z}} M_j$ then $\text{End}_{K[y^*]}(M)_k = \{\varphi: M \rightarrow M \mid \varphi(M_i) \subset M_{i+k}\}$. This equips A with an algebra grading.

It turns out there's a different interpretation of A . Namely consider the principal block $\check{\mathcal{O}}^{\text{pr}}$ for Langlands dual Lie algebra $\check{\mathfrak{g}}$. Consider the simple objects $\check{L}_0(w \cdot 0) \in \check{\mathcal{O}}^{\text{pr}}$ and the (cohomologically) graded algebra $\text{Ext}_{\check{\mathcal{O}}^{\text{pr}}}^*\left(\bigoplus_{w \in W} \check{L}_0(w \cdot 0)\right)$, where $\text{Ext}^* = \bigoplus_{i \geq 0} \text{Ext}^i$

Theorem 1 (Beilinson-Ginzburg, Soergel)

$$A \xrightarrow{\sim} \text{Ext}_{\check{\mathcal{O}}^{\text{pr}}}^* \left(\bigoplus_{w \in W} \check{L}_0(w \cdot 0) \right) \text{ as graded algebras.}$$

The proof is based on an equivalence of a suitable additive category with $\text{SMod}(W)$. The additive category of interest to be denoted by $\mathcal{D}^b(\check{\mathcal{O}}^{\text{pr}})_{ss}$. This is the full subcategory in $\mathcal{D}^b(\check{\mathcal{O}}^{\text{pr}})$ consisting of the direct sums of $\check{L}_0(w \cdot 0)[j]$, $w \in W$, $j \in \mathbb{Z}$ ($[j]$ is homological shift). The functor $\check{V}_K: M \mapsto \bigoplus_{j \in \mathbb{Z}} \text{Hom}_{\mathcal{D}^b(\check{\mathcal{O}}^{\text{pr}})}(\check{L}_0(0), M[j])$ ("K" for Koszul) take values in $\text{Ext}_{\check{\mathcal{O}}^{\text{pr}}}^*(\check{L}_0(0), \check{L}_0(0))\text{-grmod}$.

Theorem 2 (Soergel)

- 1) $\text{Ext}_{\check{\mathcal{O}}^{\text{pr}}}^*(\check{L}_0(0), \check{L}_0(0)) = \mathbb{C}[\check{J}]^{\text{cog}}$
- 2) $\check{V}_K$ is fully faithful on $\mathcal{D}^b(\check{\mathcal{O}}^{\text{pr}})_{ss}$
- 3) & $\check{V}_K(\check{L}_0(w \cdot 0)) = B_w \otimes_{K[\check{J}^*]} K_0$

Exercise: Deduce Thm 1 from Thm 2

Unlike the proofs of Soergel's theorems from Lec 1, which are algebraic and will be explained in this course, the proof of Theorem 2 is geometric. We won't cover it but we'll explain the relevant geometry.

Remark: Theorem 1 is a starting point for studying Koszul duality in Representation theory which is a very important topic.

2) Geometric interpretation

2.1) Strongly equivariant modules

We'll need a general construction. Let $\mathcal{A}$ be a unital associative algebra/ K & F be an algebraic group.

Suppose we are given an action of F on $\mathcal{A}$ by automorphisms that is **rational** meaning that $\forall a \in \mathcal{A}$

$\exists$ subspace $V \subset \mathcal{A}$ s.t. $a \in V$, $\dim V < \infty$, V is F -stable

& is an algebraic representation of F . This gives a

3) Lie algebra homomorphism $\mathfrak{f} \rightarrow \text{Der}(\mathcal{A})$, $x \mapsto x_{\mathcal{A}}$ by

differentiating the F -action.

Definition: A quantum comoment map for $F \curvearrowright \mathcal{A}$ is an F -equivariant linear map $\varphi: \mathfrak{f} \rightarrow \mathcal{A}$ s.t.

$$[\varphi(x), a] = x_{\mathcal{A}} a \quad \forall x \in \mathfrak{f}, a \in \mathcal{A}$$

It's automatically a Lie algebra homomorphism.

Example 1 a) Let G be an algebraic group, $F \subset G$ an algebraic subgroup & $\mathcal{A} = \mathcal{U}(\mathfrak{g})$, so that F naturally acts on $\mathcal{A}$. Then the composition $\varphi: \mathfrak{f} \hookrightarrow \mathfrak{g} \hookrightarrow \mathcal{U}(\mathfrak{g})$ is a quantum comoment map.

b) Let Y be a smooth affine variety. We can consider the algebra of differential operators $\mathcal{D}(Y)$ on Y . By definition,

- it contains $K[Y]$ as a subalgebra

- $\text{Vect}(Y) = \text{Der}(K[Y])$ as Lie subalgebra & left

$K[Y]$ -submodule

- the commutation between $K[Y]$ & $\text{Vect}(Y)$ is $[\xi, f] = \underline{\xi \cdot f}$ ($\xi \in \text{Vect}(Y), f \in K[Y]$).
action of ξ on $f \in K[Y]$

For example, for $Y = \mathbb{A}^1$ we get the 1st Weyl algebra $K\langle x, \partial \rangle / (\partial x - x\partial - 1)$

Now suppose F acts on Y . Applying the general construction in the beginning of the section to $F \curvearrowright K[Y]$ we get an F -equivariant map $x \mapsto x_Y := x_{K[Y]}: f \rightarrow \text{Vect}(Y)$. For $x \in f$, set $\varphi(x) = x_Y \in \text{Vect}(Y) \subset \mathcal{D}(Y)$.

Note that F acts on $\mathcal{D}(Y)$.

Exercise: φ is a quantum comoment map.

Now we return to the general situation.

Definition: A weakly F -equivariant $\mathcal{A}$ -module is an $\mathcal{A}$ -module M together w. a rational F -action s.t. the action map $\mathcal{A} \otimes M \rightarrow M$ is F -equivariant. We

5 | say M is strongly F -equivariant if $\varphi(x)m = x_M m$

$\forall x \in \mathfrak{f}, m \in M$ (where $x_M \in \text{End}(M)$ comes from differentiating the F -action). Let $\mathcal{A}\text{-mod}^F$ denote the category of strongly F -equivariant $\mathcal{A}$ -modules.

Example 2 a) Let G be a simply connected semisimple algebraic group, $B \subset G$ be a Borel subgroup. We claim that the full subcategory of fin. gen. $\mathcal{U}(\mathfrak{g})$ -modules in $\mathcal{U}(\mathfrak{g})\text{-mod}^B$ is $\mathcal{O}_0$.

First, suppose M is a strongly B -equivariant $\mathcal{U}(\mathfrak{g})$ -module. The action of max. torus $H \subset B$ gives a decomposition $M = \bigoplus_{\lambda \in \mathcal{X}(H)} M_\lambda$, where the summation is over the character lattice $\mathcal{X}(H) = \text{Hom}(H, \mathbb{C}_m) \xrightarrow{\sim} [\mathbb{C} \text{ is simply connected}] \wedge$, & $M_\lambda = \{m \in M \mid t.m = \lambda(t)m \ \forall t \in H \iff$
 $xm = \langle \lambda, x \rangle m \ \forall x \in \mathfrak{f}\}$, i.e. the same as in def'n for $\mathcal{O}_0$ strong equiv. for $H \curvearrowright M$
 (Sec 1.1 of Lec 1). The action of $\mathfrak{h}$ comes by differentiating that of N , hence locally nilpotent.

Conversely, the action of $\mathfrak{b}$ on any object $M' \in \mathcal{O}_0$ integrates to B establishing M' as object of $\mathcal{U}(\mathfrak{g})\text{-mod}^B$.

2) In the setting of Example 16, $K[Y]$ is a strongly F -equivariant $\mathcal{D}(Y)$ -module (exercise).

Remarks: 1) To recover $\mathcal{O}_X$ one replaces $\mathcal{P}$ w. $\mathcal{P} - X$, where we view X as element of $(\mathfrak{b}^*)^B$.

2) One can talk about the sheaf of differential operators $\mathcal{D}_Y$ on any smooth variety Y & the category $\mathcal{D}_Y\text{-mod}$ of sheaves of $\mathcal{D}_Y$ -modules that are quasi-coherent sheaves of $\mathcal{O}_Y$ -modules.

3) Suppose $\tilde{Y} \rightarrow Y$ be a principal F -bundle (say, locally trivial in the Zariski topology). Then the categories $\mathcal{D}_{\tilde{Y}}\text{-mod}^F$ & $\mathcal{D}_Y\text{-mod}$ are naturally equivalent (HW 1). Our main example is $\tilde{Y} = G$, a semisimple group, $F = B$ & $Y = G/B$ is the flag variety (non-affine).

2.2) From $\mathcal{D}$ -modules to category $\mathcal{O}$

Consider the action of $N \times B$ on G : $(n, b).g = ngb^{-1}$.

$\boxed{7}$ We have the following result.

Thm 1: The abelian categories $\{\text{fin. gen'd } \mathcal{D}(G) \text{ objects in } \mathcal{D}(G)\text{-mod}^{N \times B}\}$ & $\mathcal{O}^{\text{pr}}$ are equivalent.

Here's one way to see this. The category $\mathcal{D}(G)\text{-mod}^{N \times B}$ is the same as $\mathcal{D}_{G/B}\text{-mod}^N$ of strongly N -equivariant $\mathcal{D}$ -modules on G/B (cf. Remark 3). An object, say $\mathcal{F}$, in this category is, in particular, a sheaf on G/B so it makes sense to speak about its global sections $\Gamma(\mathcal{F}) \in \Gamma(\mathcal{D}_{G/B})\text{-mod}^N$ (under the equivalence w. $\mathcal{D}(G)\text{-mod}^{N \times B}$, Γ becomes taking the B -invariants).

One can describe $\Gamma(\mathcal{D}_{G/B})$ Lie theoretically. We still have the Lie algebra homomorphism $\mathfrak{g} \rightarrow \text{Vect}(G/B)$ coming from $G \curvearrowright G/B$. It gives rise to an associative algebra homomorphism

$$(1) \quad \mathcal{U}(\mathfrak{g}) \longrightarrow \Gamma(\mathcal{D}_{G/B})$$

Recall that $\mathbb{Z}$ stands for the center of $\mathcal{U}(\mathfrak{g})$.

8 | Let $m_0 = \{z \in \mathbb{Z} \mid z \text{ acts by } 0 \text{ on } K_{\text{env}}\}$. Set

$$\mathcal{U}_0(\mathfrak{g}) := \mathcal{U}(\mathfrak{g}) / \mathcal{U}(\mathfrak{g}) \mathfrak{m}_0$$

For example, for $\mathfrak{g} = \mathfrak{sl}_2$: $\mathcal{U}_0(\mathfrak{sl}_2) = \mathcal{U}(\mathfrak{sl}_2) / (ef + fe + \frac{1}{2})$

Thm 2 (Beilinson-Bernstein '81)

- i) (1) factors through $\mathcal{U}_0(\mathfrak{g}) \xrightarrow{\sim} \Gamma(\mathcal{D}_{G/B})$.
- ii) Thx to i) we have a functor

$$\Gamma: \mathcal{D}_{G/B}\text{-mod} \longrightarrow \mathcal{U}_0(\mathfrak{g})\text{-mod}$$

It's an equivalence of abelian categories.

Corollary: $\Gamma: \mathcal{D}_{G/B}\text{-mod}^N \longrightarrow \mathcal{U}_0(\mathfrak{g})\text{-mod}^N$

Let $\mathcal{O}'^{\text{pr}}$ denote the category of Noetherian objects ($\Leftrightarrow$ fin. generated over $\mathcal{U}_0(\mathfrak{g})$) in r.h.s. It does not coincide w. $\mathcal{O}^{\text{pr}}$ (the modules in both are fin. generated and have locally nilpotent action of N , but $\mathcal{O}^{\text{pr}}$ consists of modules where $\mathfrak{h}$ acts diagonalizably & $\mathfrak{m}_0$ acts locally nilpotently; while $\mathcal{O}'^{\text{pr}}$ consists of modules where $\mathfrak{m}_0$ acts by 0, while $\mathfrak{h}$ can be shown to act locally finitely)

Nevertheless, we have the following result (due to Soergel in his diploma work):

Thm 3: We have an equivalence of abelian categories

$$\mathcal{O}^{pr} \xrightarrow{\sim} \mathcal{O}'^{pr}$$

Combining Thm 3 & Corollary we basically get
Thm 1.